Quick-Start: Mission Planner and QGroundControl

Course: U21MI504 – Unmanned Aerial Vehicle Technology · **Units IV and V**
Use with: the ArduPilot simulator lab, the Unit IV build and the Unit V flight-operations session.

The figures below are **schematic layouts**, drawn to show where things are. They are not screenshots, and button names move slightly between versions. The numbered tags match the notes under each figure.

1. Which ground station?

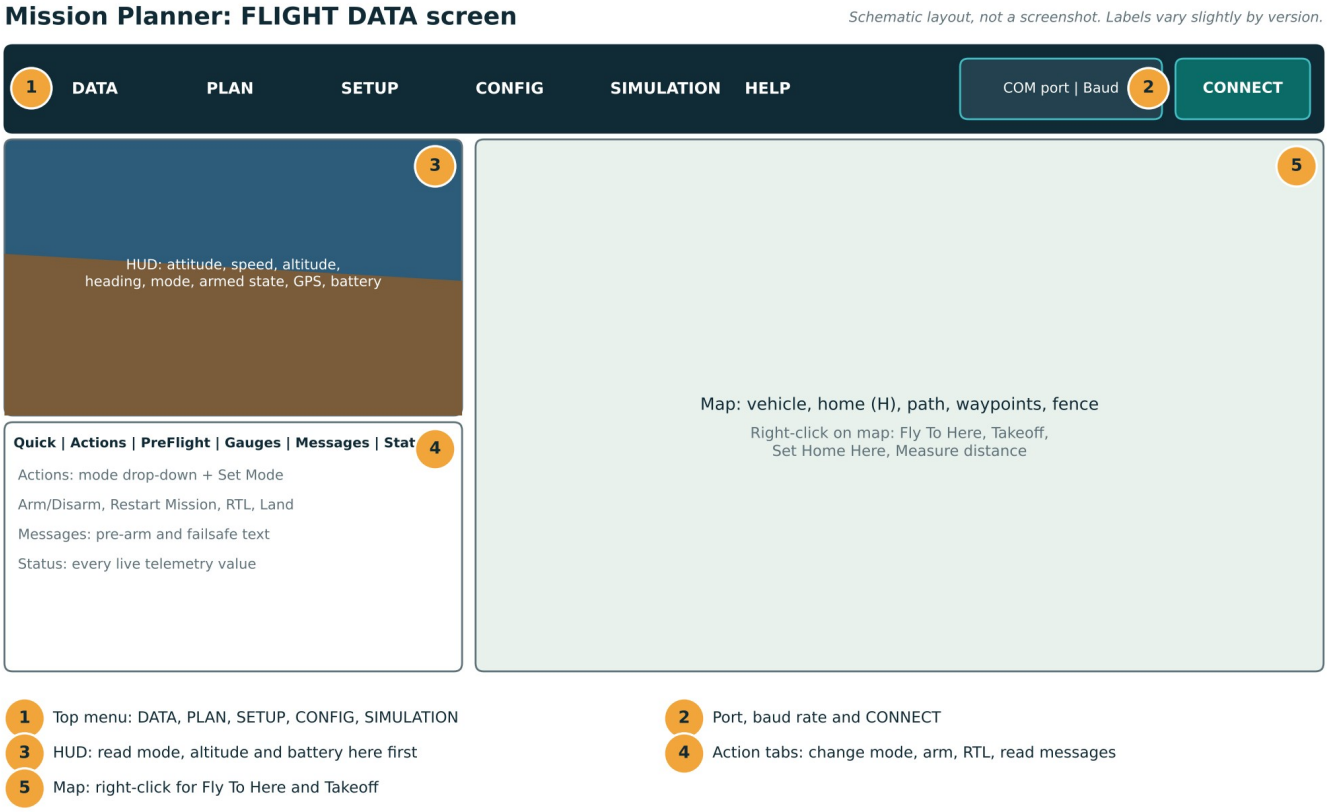
	Mission Planner	QGroundControl (QGC)
Platforms	Windows (best supported)	Windows, macOS, Linux, Android, iOS
Best with	ArduPilot	PX4 and ArduPilot
Strengths	Every parameter and log tool in one place; built-in simulator launcher	Clean touch-friendly interface; good for field tablets
Use it in this course for	SITL lab, parameter set-up, log download	Field flying on a tablet, PX4 aircraft

2. Connecting

Situation	Mission Planner	QGroundControl
USB to flight controller	Pick the COM port, baud **115200**, click CONNECT	Plug in; QGC connects automatically
Telemetry radio	Pick the radio's COM port, baud **57600** (typical), CONNECT	Plug in the radio; auto-connects
Simulator	SIMULATION tab → Multirotor	Start SITL separately; QGC connects on UDP 14550

If nothing connects: check the cable is a data cable, the driver is installed, and no other program holds the port.

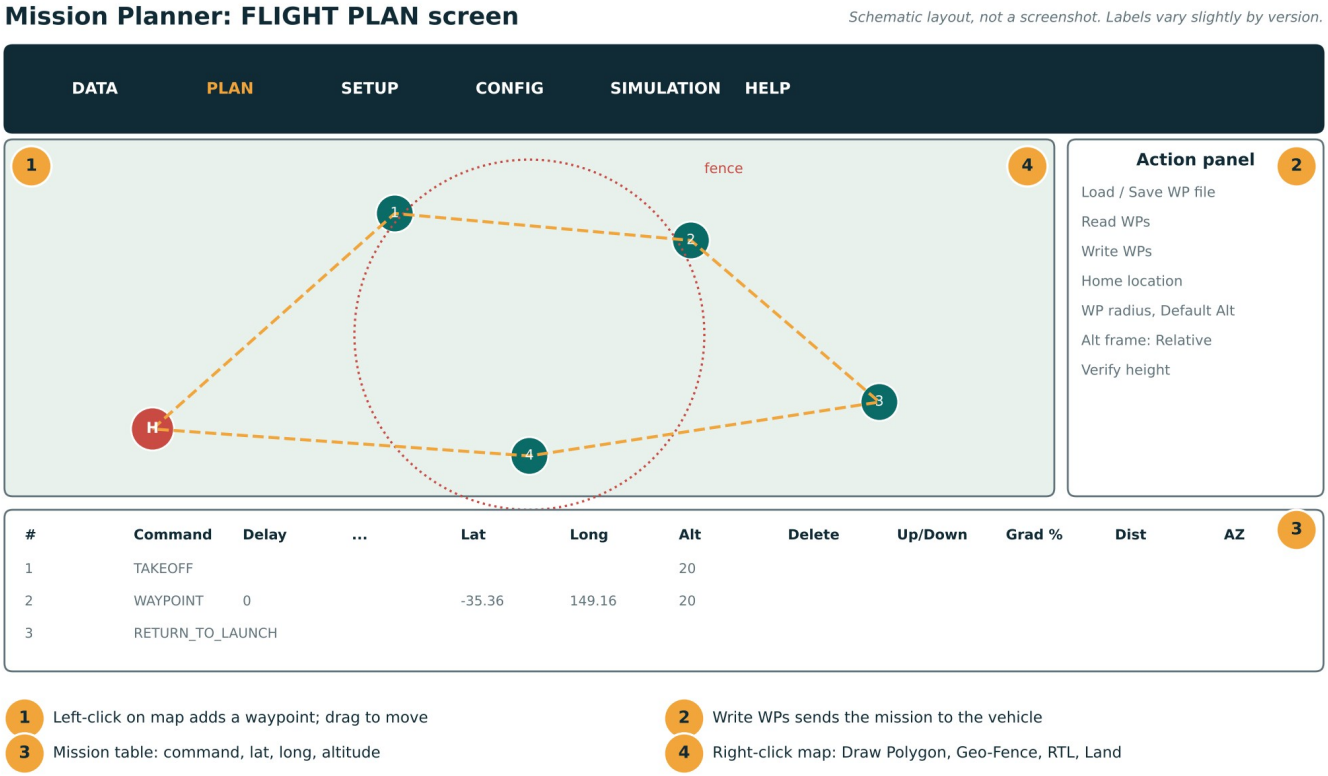
3. Mission Planner: Flight Data



Mission Planner Flight Data layout (schematic)

- Read the **HUD** before every action: mode, ARMED or DISARMED, GPS fix, battery voltage.
- **Actions** tab: choose a mode, then **Set Mode**. **Arm/Disarm** is here too.
- **Messages** tab: pre-arm failures appear here in plain text (“PreArm: GPS not healthy”, “PreArm: RC not calibrated”). Fix the cause; do not force-arm.
- Right-click on the map for **Fly To Here** (Guided) and **Takeoff**.

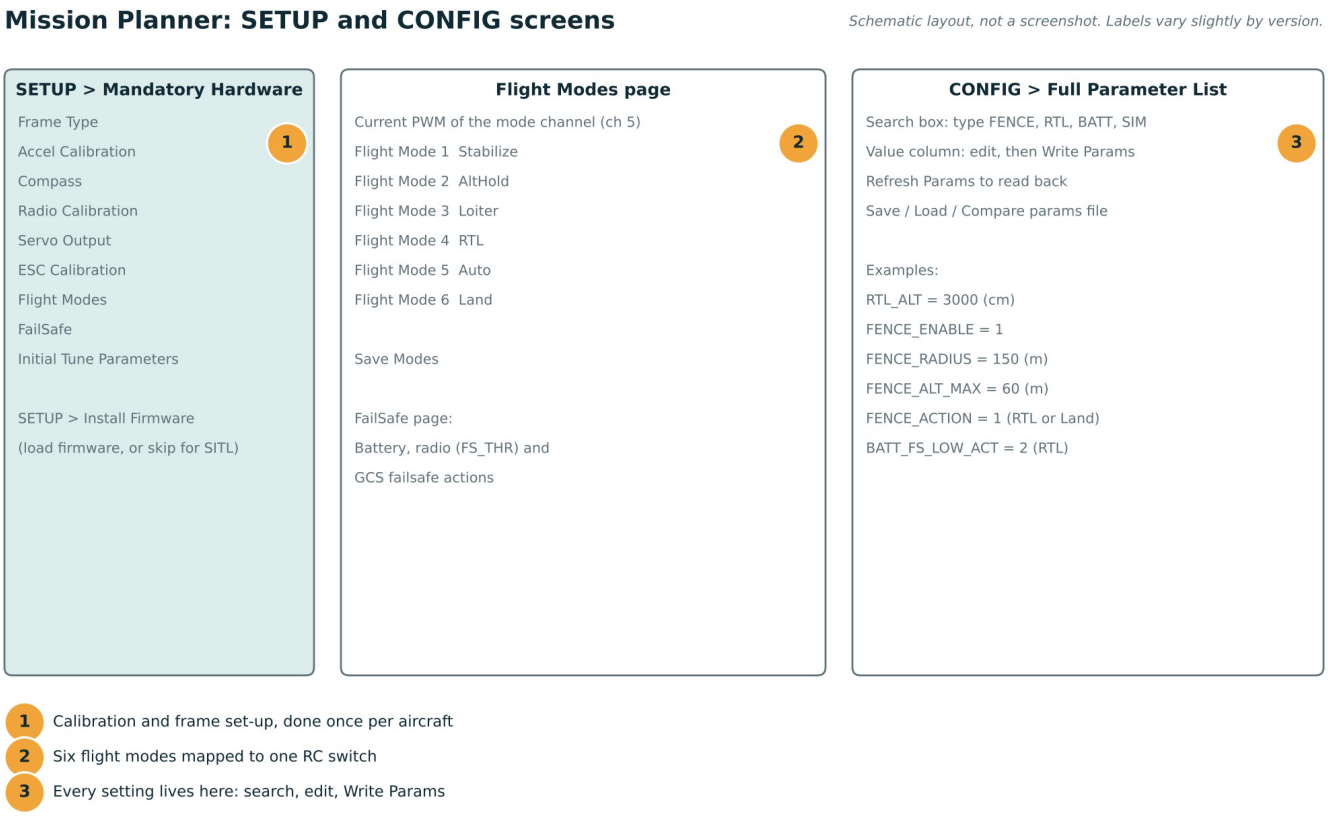
4. Mission Planner: Flight Plan



Mission Planner Flight Plan layout (schematic)

1. Set **Default Alt** and **Alt frame: Relative** (altitude above home).
2. Left-click to add waypoints. Change the first row to **TAKEOFF** and add **RETURN_TO_LAUNCH** at the end.
3. Right-click → **Geo-Fence** to draw a polygon fence if needed.
4. Click **Write WPs**, then **Read WPs** to confirm the vehicle holds what you planned.

5. Mission Planner: Setup and Config



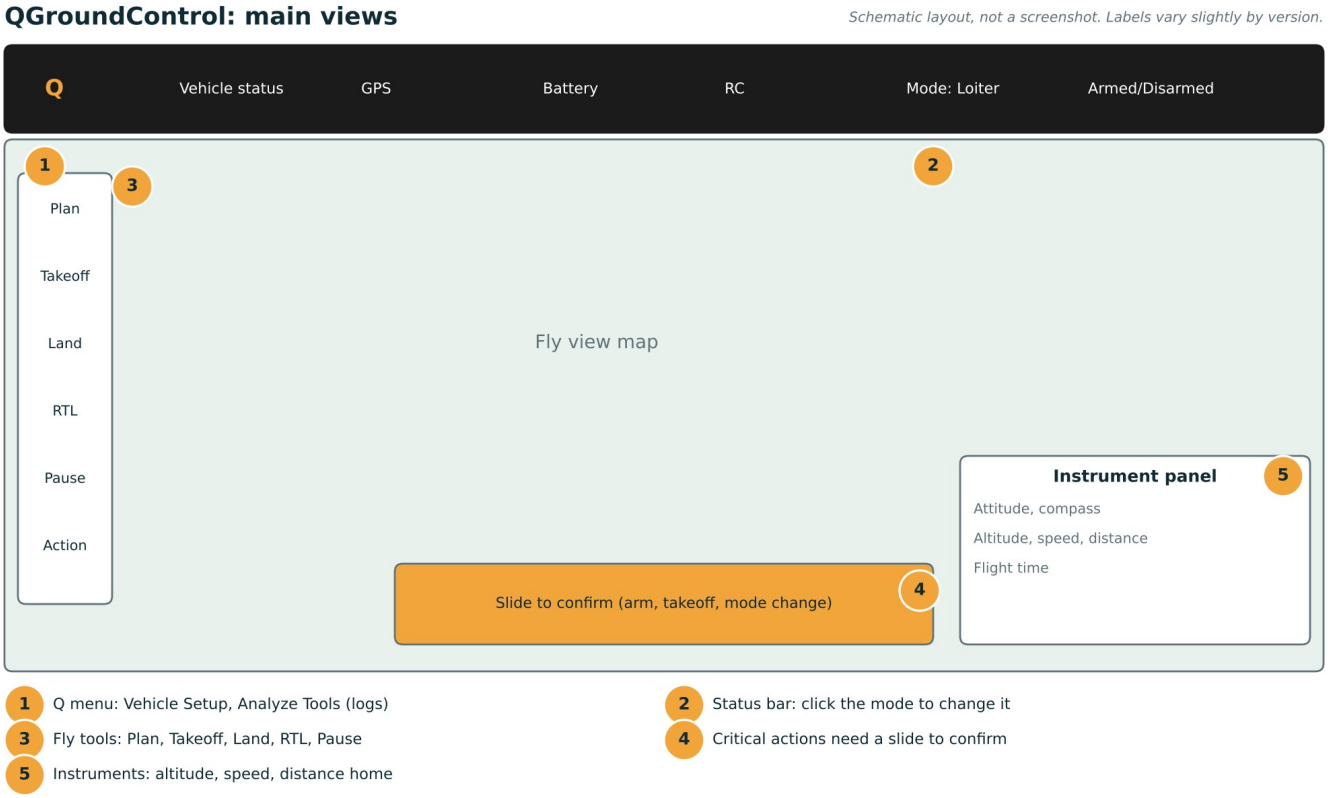
Mission Planner Setup and Config layout (schematic)

New-aircraft order (Unit IV build): **Frame Type → Accel Calibration → Compass → Radio Calibration → Servo Output → ESC Calibration → Flight Modes → FailSafe**. Props off for everything except the final hover test.

Parameter list habits:

- Search by prefix: `FENCE_`, `RTL_`, `BATT_`, `FS_`, `SIM_`.
- Click **Write Params** after editing, then **Refresh Params** to confirm.
- **Save** a parameter file before and after every change session. Name it with the date and the aircraft.
- Units differ: `RTL_ALT` is in **centimeters**, `FENCE_ALT_MAX` and `FENCE_RADIUS` are in **meters**.

6. QGroundControl



QGroundControl main views (schematic)

- **Q menu → Vehicle Setup** for the same calibration steps as Mission Planner.
- **Q menu → Analyze Tools → Log Download** to pull logs.
- Click the mode name in the top bar to change mode. Takeoff, Land and RTL need a **slide to confirm**.
- **Plan** view: add waypoints, then **Upload**.

7. Ten-second checks before you press ARM

Check	Where to look	Go if
Mode	HUD	The mode you briefed (usually Loiter or AltHold)
GPS	HUD or top bar	3D fix or better, HDOP below about 1.5 for position modes
Battery	HUD	Full and matching the pack you logged
Home	Map	“H” at your take-off point
Fence and RTL	Parameter list	Values match the site plan
Messages	Messages tab	No pre-arm errors

8. Where to go next

- ArduPilot Copter documentation (ardupilot.org/copter): flight modes, failsafes, parameter reference.
- PX4 user guide (docs.px4.io) for PX4 aircraft.
- UAV Log Viewer (plot.ardupilot.org) for log plots.